

PATRICK COWHILL

Melbourne, FL · [LinkedIn.com/in/PatrickCowhill](https://www.linkedin.com/in/PatrickCowhill) · PatrickCowhill.com · Cowhill.dev

TS/SCI-cleared Software Engineer and Technical Specialist with a Master's in Electrical and Computer Engineering (4.0 GPA) and a Bachelor's in Physics and Mathematics. Proven record of delivering mission-critical EO/SAR software and leading teams of up to 20 from research and concept to deployment. Skilled in image and signal processing and geospatial systems with experience in containerized releases, GPU acceleration, and cloud/on-prem integration.

RELEVANT EXPERIENCE

L3Harris Technologies

Melbourne, FL

Specialist (L3) Software Engineer

March 2024 - Present

Senior Associate (L2) Software Engineer

June 2022 - February 2024

Associate (L1) Software Engineer

January 2020 - May 2022

- Lead a 20-member engineering team as product owner for three deployed solutions.
- Drive \$4M in annual program revenue by leading technical demos, facilitating customer engagements, and authoring proposal volumes for innovative geospatial technologies.
- Direct development of EO and SAR software in C++, Python, and MATLAB for Linux.
- Integrate GPU acceleration to boost performance of image processing workflows.
- Containerize, test, and integrate codebases for GovCloud and on-prem deployment.
- Collaborate with end users to define requirements and guide feature growth.
- Oversee debugging, code reviews, configuration management, and stakeholder demos.

University of Florida Physics Department

Gainesville, FL

Undergraduate Research Assistant and Guest Lecturer

January 2018 - December 2019

- Modeled experimental data and ran biochemical system simulations in MATLAB.
- Designed and carried out *Vibrio fischeri* experiments for bioluminescence investigations.
- Operated lab machinery and authored protocols supporting peer-reviewed publication.
- Collaborated with Principal Investigator and graduate students on biophysics research.
- Instructed 4000-level Mechanics II, applying Mathematica to complex physics concepts.

EDUCATION

Georgia Institute of Technology

GPA: 4.00

Atlanta, GA

Master of Science in Electrical and Computer Engineering

August 2021 - May 2024

University of Florida

GPA: 3.95

Gainesville, FL

Bachelor of Science with Majors in Physics and Mathematics

August 2016 - December 2019

PATENTS

U.S. Patent No. 11,636,649 B2 – Inventor

Issued April 2023

- *Geospatial System for 3D Model Updates Based Upon Predictively Registered Image*

U.S. Patent No. 11,816,793 B2 – Inventor

Issued November 2023

- *System for 3D Geospatial Model Updates via Iterative Predictive Image Registration*

CORE COMPETENCIES

Programming & Scripting C/C++, Python, Matlab, Bash, JavaScript/HTML/CSS, Java, PowerShell

Frameworks & Libraries Numpy, SciPy, YAML, CV2, OpenGL, Qt, SFML, CUDA, REST API, VMs

DevOps & CI/CD Docker/Podman, Jenkins, AWS, Artifactory/Nexus, SonarQube, Fortify SCA

Operating Systems Red Hat Enterprise Linux 7/8, CentOS, Amazon Linux 2023, Windows 10/11

Collaboration & Program Agile, Scrum Master, Atlassian, Git (BitBucket, Github), Mentorship

Specialized Domains Image & Signal Processing, Photogrammetry, GEOINT, Data Analysis, DoD